```
Command  . . . . . . . . . . . . :   CRTBNDRPG
  Issued by  . . . . . . . . . . :     ANGO
Program  . . . . . . . . . . . . :   RPDFGCVT
  Library  . . . . . . . . . . . :     ANGO1
Text 'description' . . . . . . . :   *SRCMBRTXT
Source stream file   . . . . . . :   QRPGSRC/RPDFGCVT.SQLRPGLE
  CCSID  . . . . . . . . . . . . :     1208
Target CCSID . . . . . . . . . . :   *JOB (273)
Text 'description' . . . . . . . :
Last Change  . . . . . . . . . . :   26-04-01  16:53:30

Generation severity level  . . . :   10
Default activation group . . . . :   *YES
Compiler options . . . . . . . . :   *XREF      *GEN       *NOSECLVL  *SHOWCPY
                                     *EXPDDS    *EXT       *NOSHOWSKP *NOSRCSTMT
                                     *DEBUGIO   *UNREF     *NOEVENTF
Debugging views  . . . . . . . . :   *LIST
Debug encryption key . . . . . . :   *NONE
Output . . . . . . . . . . . . . :   *PRINT
Optimization level . . . . . . . :   *NONE
Source listing indentation . . . :   *NONE
Type conversion options  . . . . :   *NONE
Sort sequence  . . . . . . . . . :   *JOB
Language identifier  . . . . . . :   *JOB
Replace program  . . . . . . . . :   *YES
User profile . . . . . . . . . . :   *USER
Authority  . . . . . . . . . . . :   *LIBCRTAUT
Truncate numeric . . . . . . . . :   *YES
Fix numeric  . . . . . . . . . . :   *NONE
Target release . . . . . . . . . :   V7R5M0
Allow null values  . . . . . . . :   *NO
Define condition names . . . . . :   *NONE
Enable performance collection  . :   *PEP
Profiling data . . . . . . . . . :   *NOCOL
Licensed Internal Code options . :
Generate program interface . . . :   *NO
Include directory  . . . . . . . :
Preprocessor options . . . . . . :   *NORMVCOMMENT *EXPINCLUDE   *NOSEQSRC
Output source file . . . . . . . :   QSQLPRE
  Library  . . . . . . . . . . . :     QTEMP
Output source member . . . . . . :   RPDFGCVT
MINIMUM OUTPUT LINE LENGTH . . . :   100
Require prototype for export . . :   *NO
```

-=* http://pub400.com *=-

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
```

S o u r c e L i s t i n g

```
     1 **FREE                                                                                                  000001
     2 CTL-OPT MAIN(MAIN) DFTACTGRP(*NO) ACTGRP('RPDFG') BNDDIR('RPDFG')                                       000002
     3        OPTION(*NODEBUGIO:*SRCSTMT:*NOUNREF)                                                             000003
     4        DATEDIT(*YMD) CCSID(*EXACT) CCSIDCVT(*LIST:*EXCP)                                                000004
     5        TEXT('RPDFGCVT - Convert SCS spooled files to PDF format');                                      000005
     6 // ---------------------------------------------------------------------------                         000006
     7 // RPDFGCVT - Convert SCS spooled files to PDF format                                                   000007
     8 // ---------------------------------------------------------------------------                         000008
     9 Dcl-PR MAIN EXTPGM('RPDFGCVT');                                                                         000009
    10    SplfName        CHAR( 10) CONST;                                                                     000010
    11    QualJob         CHAR( 26) CONST;                                                                     000011
    12    SplfNbr          INT( 10) CONST;                                                                     000012
    13    PDF_Path     VARCHAR(128) CONST;                                                                     000013
    14    PDF_File     VARCHAR(128) CONST;                                                                     000014
    15    PaperFormat  VARCHAR( 20) CONST;                                                                     000015
    16    PageRotation    CHAR( 10) CONST;                                                                     000016
    17    Margins       LIKEDS(RPDFG_Margins_T)  CONST;                                                        000017
    18    TimeZones     LIKEDS(RPDFG_TimeZone_T) CONST;                                                        000018
    19    Options       LIKEDS(RPDFG_Options_T)  CONST;                                                        000019
    20 End-PR;                                                                                                 000020
    21                                                                                                         000021
    22 // max. number of "Pages" objects in the "Pages" tree                                                   000022
    23 Dcl-C  PAGES_PER_PAGES        5;                                                                       000023
    24 // max. number of "Page" objects in a "Pages" object                                                    000024
    25 Dcl-C  PAGE_PER_PAGES         5;                                                                       000025
    26                                                                                                         000026
    27 Dcl-C  SQL_OK                 0;                                                                       000027
    28 Dcl-C  SQL_ERROR_NOTFOUND     100;                                                                     000028
    29                                                                                                         000029
    30 Dcl-C  SPLFNBR_ONLY_NUM       0;                                                                       000030
    31 Dcl-C  SPLFNBR_LAST_NUM       -1;                                                                      000031
    32                                                                                                         000032
    33 Dcl-DS CCSID_JOBRUN CONST QUALIFIED;                                                                    000033
    34    SLASH       CHAR(1) INZ('/') CCSID(*JOBRUN);                                                         000034
    35    COLON       CHAR(1) INZ(':') CCSID(*JOBRUN);                                                         000035
    36    UNDERLINE   CHAR(1) INZ('_') CCSID(*JOBRUN);                                                         000036
    37 End-DS;                                                                                                 000037
    38                                                                                                         000038
    39 Dcl-S  RPDFGMSGF  CHAR(21) CONST CCSID(*JOBRUN) INZ('*LIBL/RPDFGMSGF');                                 000039
    40 Dcl-DS MSGID QUALIFIED CONST;                                                                           000040
    41    RPDF001     CHAR(7) INZ('RPDF001') CCSID(*JOBRUN);                                                   000041
    42    RPDF003     CHAR(7) INZ('RPDF003') CCSID(*JOBRUN);                                                   000042
    43    RPDF004     CHAR(7) INZ('RPDF004') CCSID(*JOBRUN);                                                   000043
    44    RPDF005     CHAR(7) INZ('RPDF005') CCSID(*JOBRUN);                                                   000044
    45    RPDF006     CHAR(7) INZ('RPDF006') CCSID(*JOBRUN);                                                   000045
    46    RPDF007     CHAR(7) INZ('RPDF007') CCSID(*JOBRUN);                                                   000046
    47    RPDF008     CHAR(7) INZ('RPDF008') CCSID(*JOBRUN);                                                   000047
    48    RPDF009     CHAR(7) INZ('RPDF009') CCSID(*JOBRUN);                                                   000048
    49 End-DS;                                                                                                 000049
    50                                                                                                         000050
    51 Dcl-ENum PAGRTT QUALIFIED;                                                                              000051
    52    AUTO              '*AUTO';                                                                           000052
    53    DEVD              '*DEVD';                                                                           000053
```

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Line   <---------------------- Source Specifications ----------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
    54    COR               '*COR';                                                                                   000054
    55    PORTRAIT          '0';                                                                                      000055
    56    LANDSCAPE         '90';                                                                                     000056
    57    PORTRAIT_REVERSE  '180';                                                                                    000057
    58    LANDSCAPE_REVERSE '270';                                                                                    000058
    59 End-ENum;                                                                                                      000059
    60                                                                                                                000060
    61 Dcl-ENum OPTIONS QUALIFIED;                                                                                    000061
    62    NONE              '*NONE';                                                                                  000062
    63    JOBINFO           '*JOBINFO';                                                                               000063
    64    NUMBERS           '*NUMBERS';                                                                               000064
    65    GREENBAR          '*GREENBAR';                                                                              000065
    66 End-ENum;                                                                                                      000066
    67                                                                                                                000067
    68 Dcl-PR QCMDEXC EXTPGM('QCMDEXC');                                                                              000068
    69    CmdString     CHAR(32702) CONST OPTIONS(*VARSIZE);                                                          000069
    70    CmdLength   PACKED(15:5)  CONST;                                                                            000070
    71 End-PR;                                                                                                        000071
    72                                                                                                                000072
    73 // ------------------------------------------------------------------------                                   000073
    74 Dcl-DS SQL_Diag QUALIFIED;                                                                                     000074
    75    MessageId        char(10) ;                                                                                 000075
    76    MessageId1       varchar(7) ;                                                                               000076
    77    MessageId2       like(SQL_Diag.MessageId1) ;                                                                000077
    78    MessageLength    int(5) ;                                                                                   000078
    79    MessageText      varchar(32740) ;                                                                           000079
    80    ReturnedSQLCode  char(5) ;                                                                                  000080
    81    ReturnedSQLState char(5) ;                                                                                  000081
    82    RowsCount        int(10) ;                                                                                  000082
    83 End-DS;                                                                                                        000083
    84                                                                                                                000084
    85 // Program Status Data Structure                                                                              000085
    86 Dcl-DS PSDS PSDS QUALIFIED;                                                                                    000086
    87    JobName   CHAR(10) POS(244);                                                                                000087
    88    JobUser   CHAR(10) POS(254);                                                                                000088
    89    JobNumber CHAR( 6) POS(264);                                                                                000089
    90    CompileUDate CHAR( 6) POS(288);                                                                             000090
    91    CompileTime  CHAR( 6) POS(294);                                                                             000091
    92 End-DS;                                                                                                        000092
    93                                                                                                                000093
    94 // message data for message ids RPDF0002 - RPDF0009                                                           000094
    95 Dcl-DS RPDF000n QUALIFIED;                                                                                     000095
    96    idx        INT(10);                                                                                         000096
    97    maxIdx     INT(10);                                                                                         000097
    98    key        CHAR(16);                                                                                        000098
    99    value1     CHAR(128);                                                                                       000099
   100    value2     CHAR(128);                                                                                       000100
   101 End-DS;                                                                                                        000101
   102                                                                                                                000102
   103 /INCLUDE 'QSRVCPY/RPDFG.RPGLEINC'                                                                              000103


Line   <---------------------- Source Specifications ---------------------------><---- Comments ----> Do  Page  Change Src Seq
Number ....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10 Num Line  Date   Id  Number
        *------------------------------------------------------------------------------------------*
        * RPG member name  . . . . . :  RPDFG.RPG+                                                 *                     1
```

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Line   <---------------------- Source Specifications --------------------------><---- Comments ----> Do  Page  Change Src Seq
Number ....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10 Num Line  Date   Id  Number
        * External name  . . . . . . :  ./QSRVCPY/RPDFG.RPGLEINC                                    *                      1
        * Last change  . . . . . . . :  26-04-01  10:48:01                                          *                      1
        *-----------------------------------------------------------------------------------------*


Line   <---------------------- Source Specifications ----------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   104+**FREE                                                                                                                1000001
   105+// Prototypings and templates for the RPDFG service program                                                           1000002
   106+                                                                                                                        1000003
   107+Dcl-S  RPDFG_objIdx_T         INT(10) TEMPLATE;                                                                        1000004
   108+Dcl-S  RPDFG_dictIdx_T        INT(10) TEMPLATE;                                                                        1000005
   109+Dcl-S  RPDFG_32K_StringUTF8_T VARCHAR(32000) CCSID(*UTF8) TEMPLATE;                                                    1000006
   110+                                                                                                                        1000007
   111+Dcl-S  RPDFG_FontSize_T       PACKED(8:3) INZ(0);                                                                      1000008
   112+                                                                                                                        1000009
   113+Dcl-DS RPDFG_SplfAtr_T QUALIFIED TEMPLATE;                                                                              1000010
   114+   OutqName        VARCHAR(10);                                                                                         1000011
   115+   OutqLib         VARCHAR(10);                                                                                         1000012
   116+   SplfName        VARCHAR(10);                                                                                         1000013
   117+   SplfNbr         INT(10);                                                                                             1000014
   118+   UserName        VARCHAR(10);                                                                                         1000015
   119+   UserData        VARCHAR(10);                                                                                         1000016
   120+   Job_Name        VARCHAR(28);                                                                                         1000017
   121+   CreateTimestamp TIMESTAMP(0);                                                                                         1000018
   122+   Total_Pages     INT(10);                                                                                             1000019
   123+   FormType        VARCHAR(10);                                                                                         1000020
   124+   Page_Length     INT(10);                                                                                             1000021
   125+   Page_Width      INT(10);                                                                                             1000022
   126+   Page_Rotation   VARCHAR(5);                                                                                          1000023
   127+End-DS;                                                                                                                 1000024
   128+                                                                                                                        1000025
   129+Dcl-DS RPDFG_Margins_T QUALIFIED TEMPLATE;                                                                              1000026
   130+   NbrOfElems       INT(5);                                                                                              1000027
   131+   // all margins in "points" (= 1/72 inch)                                                                              1000028
   132+   Left          PACKED(8:3);                                                                                           1000029
   133+   Top           PACKED(8:3);                                                                                           1000030
   134+   Right         PACKED(8:3);                                                                                           1000031
   135+   Bottom        PACKED(8:3);                                                                                           1000032
   136+End-DS;                                                                                                                 1000033
   137+                                                                                                                        1000034
   138+Dcl-DS RPDFG_TimeZone_T QUALIFIED TEMPLATE;                                                                             1000035
   139+   TimeZoneNormal     CHAR(5);                                                                                           1000036
   140+   TimeZoneSummer     CHAR(5);                                                                                           1000037
   141+   SummerStartMonth   PACKED(2:0);                                                                                       1000038
   142+   SummerStartSunday  CHAR(1);                                                                                           1000039
   143+   SummerEndMonth     PACKED(2:0);                                                                                       1000040
   144+   SummerEndSunday    CHAR(1);                                                                                           1000041
   145+End-DS;                                                                                                                 1000042
   146+                                                                                                                        1000043
   147+Dcl-DS RPDFG_Options_T QUALIFIED TEMPLATE;                                                                              1000044
   148+   NbrOfElems       INT(5);                                                                                              1000045
   149+   value           CHAR(10) DIM(10);                                                                                     1000046
   150+End-DS;                                                                                                                 1000047
   151+                                                                                                                        1000048
   152+Dcl-DS RPDFG_obj_T QUALIFIED TEMPLATE;                                                                                  1000049
```

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Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   153+   inUse                  IND INZ(*OFF);                                                                          1000050
   154+   objNumberForPDF    INT(10) INZ(-1); // this is different from the index!                                       1000051
   155+   SpecialMode        CHAR(1) INZ(RPDFG_SPECIAL_OBJECT.NONE);                                                     1000052
   156+   isArray            CHAR(1) INZ(*OFF);                                                                          1000053
   157+   Parent             LIKE(RPDFG_ref) INZ('');                                                                    1000054
   158+   Kids               LIKE(RPDFG_ref) INZ('') DIM(50);                                                            1000055
   159+   NbrOfKids          INT(10) INZ(0);                                                                             1000056
   160+   dictIdx            INT(10) INZ(-1);                                                                            1000057
   161+   // PtrDict            POINTER INZ(*NULL);                                                                       1000058
   162+   // PtrStream          POINTER INZ(*NULL);                                                                       1000059
   163+   // Content       VARCHAR(1000);                                                                                 1000060
   164+   // ContentSep    VARCHAR(1);                                                                                    1000061
   165+End-DS;                                                                                                           1000062
   166+                                                                                                                  1000063
   167+Dcl-DS RPDFG_dict_T QUALIFIED TEMPLATE;                                                                           1000064
   168+   inUse                  IND INZ(*OFF);                                                                          1000065
   169+   NbrOfDictEntries   INT(10) INZ(0);                                                                            1000066
   170+   DictKeys     VARCHAR(  16) DIM(32) INZ('');                                                                   1000067
   171+   DictValues   VARCHAR(1024) DIM(32) INZ('');                                                                   1000068
   172+   PtrDictInDict      POINTER INZ(*NULL);                                                                         1000069
   173+End-DS;                                                                                                           1000070
   174+                                                                                                                  1000071
   175+Dcl-ENum RPDFG_SPECIAL_OBJECT QUALIFIED;                                                                          1000072
   176+   NONE        ' ';                                                                                               1000073
   177+   CONTENTS    'C';                                                                                               1000074
   178+   INTERNAL    'I';                                                                                               1000075
   179+   TRAILER     'T';                                                                                               1000076
   180+   XREF        'X';                                                                                               1000077
   181+End-ENum;                                                                                                         1000078
   182+                                                                                                                  1000079
   183+// ---------------------------------------------------------------------------                                   1000080
   184+// Set paper format                                                                                               1000081
   185+// - paperFormat must be one of                                                                                   1000082
   186+//   [A3,A4,A5,                                                                                                    1000083
   187+//    US INVOICE,US EXECUTIVE,US LEGAL,US LETTER,US LEDGER]                                                        1000084
   188+Dcl-PR RPDFG_initPaperFormat  EXTPROC(*DCLCASE);                                                                  1000085
   189+   paperFormat      VARCHAR(20)  CONST;                                                                           1000086
   190+   // Margins in "points" (= 1/72 inch)                                                                           1000087
   191+   MarginLeftPts    PACKED(8:3) CONST;                                                                            1000088
   192+   MarginTopPts     PACKED(8:3) CONST;                                                                            1000089
   193+   MarginRightPts   PACKED(8:3) CONST;                                                                            1000090
   194+   MarginBottomPts  PACKED(8:3) CONST;                                                                            1000091
   195+End-PR;                                                                                                           1000092
   196+                                                                                                                  1000093
   197+Dcl-PR RPDFG_calculatePrintingValues EXTPROC(*DCLCASE);                                                           1000094
   198+   Override_Page_Rotation    CHAR(10) CONST;                                                                      1000095
   199+   Page_Rotation             LIKE(RPDFG_SplfAtr_T.Page_Rotation) CONST;                                           1000096
   200+   Page_Length               LIKE(RPDFG_SplfAtr_T.Page_Length)   CONST;                                           1000097
   201+   Page_Width                LIKE(RPDFG_SplfAtr_T.Page_Width)    CONST;                                           1000098
   202+End-PR;                                                                                                           1000099
   203+                                                                                                                  1000100
   204+// get the font size                                                                                              1000101
   205+Dcl-PR RPDFG_getFontSize LIKE(RPDFG_FontSize_T) EXTPROC(*DCLCASE);                                                1000102
   206+End-PR;                                                                                                           1000103
   207+                                                                                                                  1000104
   208+// get the media box string, e.g. "[0 0 595 842]"                                                                 1000105
```

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Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   209+Dcl-PR RPDFG_getMediaBoxString  VARCHAR(32) CCSID(*UTF8) EXTPROC(*DCLCASE);                                    1000106
   210+End-PR;                                                                                                         1000107
   211+                                                                                                                1000108
   212+// get the rotation of the page (0, 90, 180, 270)                                                               1000109
   213+Dcl-PR RPDFG_getRotate INT(5) EXTPROC(*DCLCASE);                                                                1000110
   214+End-PR;                                                                                                         1000111
   215+                                                                                                                1000112
   216+// create a new PDF object and return the index to it                                                           1000113
   217+// -  specialMode 'X' for 'xref' object                                                                         1000114
   218+Dcl-PR RPDFG_newObject     LIKE(RPDFG_objIdx_T) EXTPROC(*DCLCASE);                                             1000115
   219+   specialMode  CHAR(1) CONST OPTIONS(*NOPASS);                                                                 1000116
   220+End-PR;                                                                                                         1000117
   221+                                                                                                                1000118
   222+// create a new PDF dictionary and return the index to it                                                       1000119
   223+Dcl-PR RPDFG_newDictionary LIKE(RPDFG_dictIdx_T) EXTPROC(*DCLCASE);                                             1000120
   224+End-PR;                                                                                                         1000121
   225+                                                                                                                1000122
   226+Dcl-PR RPDFG_rtvPtrObj     POINTER EXTPROC(*DCLCASE);                                                           1000123
   227+   objIdx       LIKE(RPDFG_objIdx_T) CONST;                                                                     1000124
   228+End-PR;                                                                                                         1000125
   229+                                                                                                                1000126
   230+Dcl-PR RPDFG_rtvPtrDict    POINTER EXTPROC(*DCLCASE);                                                           1000127
   231+   objIdx       LIKE(RPDFG_objIdx_T) CONST;                                                                     1000128
   232+End-PR;                                                                                                         1000129
   233+                                                                                                                1000130
   234+// retrieve a reference to an object number for PDF output                                                      1000131
   235+Dcl-PR RPDFG_ref           VARCHAR(32) EXTPROC(*DCLCASE);                                                      1000132
   236+   objIdx       LIKE(RPDFG_objIdx_T) CONST;                                                                     1000133
   237+End-PR;                                                                                                         1000134
   238+                                                                                                                1000135
   239+// get the dictionary index of an object                                                                         1000136
   240+Dcl-PR RPDFG_getObjDictIdx LIKE(RPDFG_dictIdx_T) EXTPROC(*DCLCASE);                                            1000137
   241+   objIdx       LIKE(RPDFG_objIdx_T) CONST;                                                                     1000138
   242+End-PR;                                                                                                         1000139
   243+                                                                                                                1000140
   244+// get the number of kids of an object                                                                           1000141
   245+Dcl-PR RPDFG_getNumberOfKids INT(10) EXTPROC(*DCLCASE);                                                         1000142
   246+   objIdx       LIKE(RPDFG_objIdx_T) CONST;                                                                     1000143
   247+End-PR;                                                                                                         1000144
   248+                                                                                                                1000145
   249+// get the total number of objects (for the trailer)                                                            1000146
   250+Dcl-PR RPDFG_getTotalNumberOfObjects INT(10) EXTPROC(*DCLCASE);                                                1000147
   251+End-PR;                                                                                                         1000148
   252+                                                                                                                1000149
   253+// set the parent:kids relations in the Pages tree                                                              1000150
   254+Dcl-PR RPDFG_setRelation   EXTPROC(*DCLCASE);                                                                   1000151
   255+   parent_objIdx  LIKE(RPDFG_objIdx_T) CONST;                                                                   1000152
   256+   kidobjIdx      LIKE(RPDFG_objIdx_T) CONST;                                                                   1000153
   257+End-PR;                                                                                                         1000154
   258+                                                                                                                1000155
   259+// add an entry to a dictionary (and link the dictionary to the object)                                         1000156
   260+Dcl-PR RPDFG_addDictEntry  EXTPROC(*DCLCASE);                                                                   1000157
   261+   objIdx       LIKE(RPDFG_objIdx_T) CONST;                                                                     1000158
   262+   key          VARCHAR(  16) CONST;                                                                            1000159
   263+   value        VARCHAR(1024) CONST;                                                                            1000160
   264+End-PR;                                                                                                         1000161
```

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Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   265+                                                                                                                    1000162
   266+// Serialize an object as a string and write it in the IFS                                                          1000163
   267+// Attention: after executing, the object index is no longer valid!                                                 1000164
   268+Dcl-PR RPDFG_serializeObject  EXTPROC(*DCLCASE);                                                                     1000165
   269+   objIdx       LIKE(RPDFG_objIdx_T) CONST;                                                                          1000166
   270+End-PR;                                                                                                             1000167
   271+                                                                                                                    1000168
   272+// Delete an object (mark it as not in use)                                                                           1000169
   273+Dcl-PR RPDFG_deleteObject  EXTPROC(*DCLCASE);                                                                        1000170
   274+   objIdx       LIKE(RPDFG_objIdx_T) CONST;                                                                          1000171
   275+End-PR;                                                                                                             1000172
   276+                                                                                                                    1000173
   277+// Write yet unwritten objects to the IFS (before closing the file)                                                  1000174
   278+Dcl-PR RPDFG_serializeUnwrittenObjects  EXTPROC(*DCLCASE);                                                           1000175
   279+End-PR;                                                                                                             1000176
   280+                                                                                                                    1000177
   281+// Serialize a dictionary and return the contents as an UTF-8 string                                                 1000178
   282+Dcl-PR RPDFG_serializeDict LIKE(RPDFG_32K_StringUTF8_T) RTNPARM EXTPROC(*DCLCASE);                                  1000179
   283+   dictIdx      LIKE(RPDFG_dictIdx_T) CONST;                                                                         1000180
   284+End-PR;                                                                                                             1000181
   285+                                                                                                                    1000182
   286+// Calculate the position of the line                                                                                 1000183
   287+Dcl-PR RPDFG_calcLinePos   EXTPROC(*DCLCASE);                                                                        1000184
   288+   // line number of the SCS spooled file                                                                            1000185
   289+   lineNbr      INT(10)   CONST;                                                                                     1000186
   290+   // character position (normally 1)                                                                               1000187
   291+   charPos      INT(10)   CONST;                                                                                     1000188
   292+   // returned position in points (1/72 inch) from the left edge of the page                                         1000189
   293+   posX         PACKED(8:3);                                                                                         1000190
   294+   // returned position in points (1/72 inch) from the bottom edge of the page                                       1000191
   295+   posY         PACKED(8:3);                                                                                         1000192
   296+End-PR;                                                                                                             1000193
   297+                                                                                                                    1000194
   298+// start the stream of a page object                                                                                  1000195
   299+Dcl-PR RPDFG_startStream   EXTPROC(*DCLCASE);                                                                        1000196
   300+End-PR;                                                                                                             1000197
   301+                                                                                                                    1000198
   302+// add UTF8 parameter value to the stream of a page object                                                            1000199
   303+Dcl-PR RPDFG_addToStreamUTF8   EXTPROC(*DCLCASE);                                                                    1000200
   304+   string       VARCHAR(1024)      CONST OPTIONS(*TRIM) CCSID(*UTF8);                                                1000201
   305+End-PR;                                                                                                             1000202
   306+                                                                                                                    1000203
   307+// add hex value via pointer to the stream of a page object                                                           1000204
   308+Dcl-PR RPDFG_addToStreamHex   EXTPROC(*DCLCASE);                                                                     1000205
   309+   Ptr_BufferToWrite       POINTER CONST;                                                                           1000206
   310+   Len_BufferToWrite       INT(10) CONST;                                                                           1000207
   311+End-PR;                                                                                                             1000208
   312+                                                                                                                    1000209
   313+// end the stream of a page object                                                                                    1000210
   314+Dcl-PR RPDFG_endStream     EXTPROC(*DCLCASE);                                                                       1000211
   315+End-PR;                                                                                                             1000212
   316+                                                                                                                    1000213
   317+// add a text block to the stream of a page object                                                                    1000214
   318+Dcl-PR RPDFG_addToStreamTextBlock   EXTPROC(*DCLCASE);                                                               1000215
   319+   // line number   (1 - 255)                                                                                        1000216
   320+   lineNumber       INT(10)        CONST;                                                                            1000217
```

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   321+   // column number (1 - 378)                                                                                  1000218
   322+   column           INT(10)        CONST;                                                                       1000219
   323+   // line to write                                                                                             1000220
   324+   string       VARCHAR(378)       CONST;                                                                       1000221
   325+End-PR;                                                                                                        1000222
   326+                                                                                                               1000223
   327+// add a text block to the stream of a page object                                                             1000224
   328+Dcl-PR RPDFG_addToStreamTextBlock1252   EXTPROC(*DCLCASE);                                                     1000225
   329+   // line number   (1 - 255)                                                                                   1000226
   330+   lineNumber       INT(10)        CONST;                                                                       1000227
   331+   // column number (1 - 378)                                                                                  1000228
   332+   column           INT(10)        CONST;                                                                       1000229
   333+   // line to write                                                                                             1000230
   334+   string       VARCHAR(378)       CONST;                                                                       1000231
   335+End-PR;                                                                                                        1000232
   336+                                                                                                               1000233
   337+// add a UCS2 text block to the stream of a page object                                                         1000234
   338+Dcl-PR RPDFG_addToStreamTextBlockUcs2   EXTPROC(*DCLCASE);                                                     1000235
   339+   // line number   (1 - 255)                                                                                   1000236
   340+   lineNumber       INT(10)        CONST;                                                                       1000237
   341+   // column number (1 - 378)                                                                                  1000238
   342+   column           INT(10)        CONST;                                                                       1000239
   343+   // line to write                                                                                             1000240
   344+   string       VARUCS2(378)       CONST;                                                                       1000241
   345+End-PR;                                                                                                        1000242
   346+                                                                                                               1000243
   347+                                                                                                               1000244
   348+                                                                                                               1000245
   349+// Open the IFS file for output                                                                                1000246
   350+Dcl-PR RPDFG_openPdfFile   EXTPROC(*DCLCASE);                                                                  1000247
   351+   IFSPath       VARCHAR(128) CONST OPTIONS(*TRIM);                                                            1000248
   352+   IFSFileName   VARCHAR(128) CONST OPTIONS(*TRIM);                                                            1000249
   353+   SplfAtr        LIKEDS(RPDFG_SplfAtr_T) CONST;                                                               1000250
   354+End-PR;                                                                                                        1000251
   355+                                                                                                               1000252
   356+// write something to the IFS                                                                                  1000253
   357+Dcl-PR RPDFG_writeBuffer   EXTPROC(*DCLCASE);                                                                  1000254
   358+   Ptr_Buffer    POINTER      CONST;                                                                           1000255
   359+   Buffer_Len    INT(10)      CONST;                                                                           1000256
   360+End-PR;                                                                                                        1000257
   361+                                                                                                               1000258
   362+// Close the IFS file                                                                                          1000259
   363+Dcl-PR RPDFG_closePdfFile  EXTPROC(*DCLCASE);                                                                  1000260
   364+End-PR;                                                                                                        1000261
   365+                                                                                                               1000262
   366+                                                                                                               1000263
   367+// Jobinfo at the bottom                                                                                       1000264
   368+Dcl-PR RPDFG_createPrePrintForm_JobInfo EXTPROC(*DCLCASE);                                                     1000265
   369+   Ptr_Buffer    POINTER      CONST;                                                                           1000266
   370+End-PR;                                                                                                        1000267
   371+                                                                                                               1000268
   372+// Create small numbers on the left and on the right of the page (for debugging purposes)                      1000269
   373+Dcl-PR RPDFG_createPrePrintForm_Numbers EXTPROC(*DCLCASE);                                                     1000270
   374+   Ptr_Buffer    POINTER      CONST;                                                                           1000271
   375+End-PR;                                                                                                        1000272
   376                                                                                                                 000104
```

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Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   377 /INCLUDE 'QSYSCPY/DIFS.RPGLEINC'                                                                          000105


Line   <---------------------- Source Specifications ---------------------------><---- Comments ----> Do  Page  Change Src Seq
Number ....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10 Num Line  Date   Id  Number
        *-----------------------------------------------------------------------------------------*
        * RPG member name  . . . . . :  DIFS.RPGL+                                                *                     2
        * External name  . . . . . . :  ./QSYSCPY/DIFS.RPGLEINC                                    *                     2
        * Last change  . . . . . . . :  26-03-27  08:21:08                                         *                     2
        *-----------------------------------------------------------------------------------------*


Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   378+**FREE                                                                                                  2000001
   379+dcl-pr IFS_Open int(10) EXTPROC('open');                                                                2000002
   380+   FileName pointer VALUE OPTIONS(*STRING);                                                             2000003
   381+   AccessMode uns(10) VALUE;                                                                            2000004
   382+   Mode uns(10) VALUE OPTIONS(*NOPASS);                                                                 2000005
   383+   CodepageFile uns(10) VALUE OPTIONS(*NOPASS);                                                          2000006
   384+   CodePageFrom uns(10) VALUE OPTIONS(*NOPASS);                                                          2000007
   385+end-pr IFS_Open;                                                                                        2000008
   386+                                                                                                        2000009
   387+dcl-pr IFS_Close int(10) EXTPROC('close');                                                              2000010
   388+   handle int(10) VALUE;                                                                                2000011
   389+end-pr IFS_Close;                                                                                       2000012
   390+                                                                                                        2000013
   391+dcl-pr IFS_Write int(10) EXTPROC('write');                                                              2000014
   392+   handle int(10) VALUE;                                                                                2000015
   393+   Buffer pointer VALUE;                                                                               2000016
   394+   Size int(10) VALUE;                                                                                  2000017
   395+end-pr IFS_Write;                                                                                       2000018
   396+                                                                                                        2000019
   397+dcl-pr IFS_Read int(10) EXTPROC('read');                                                                2000020
   398+   handle int(10) VALUE;                                                                                2000021
   399+   Buffer pointer VALUE;                                                                               2000022
   400+   Size int(10) VALUE;                                                                                  2000023
   401+end-pr IFS_Read;                                                                                        2000024
   402+                                                                                                        2000025
   403+dcl-pr IFS_Stat int(10) EXTPROC('stat');                                                                2000026
   404+   FileName pointer VALUE OPTIONS(*STRING);                                                             2000027
   405+   OutputDS pointer VALUE;                                                                              2000028
   406+end-pr IFS_Stat;                                                                                        2000029
   407+                                                                                                        2000030
   408+dcl-pr IFS_OpenDir pointer EXTPROC('opendir');                                                          2000031
   409+   Path pointer VALUE OPTIONS(*STRING);                                                                2000032
   410+end-pr IFS_OpenDir;                                                                                     2000033
   411+                                                                                                        2000034
   412+dcl-pr IFS_ReadDir pointer EXTPROC('readdir');                                                          2000035
   413+   DirStructure pointer VALUE;                                                                         2000036
   414+end-pr IFS_ReadDir;                                                                                     2000037
   415+                                                                                                        2000038
   416+dcl-pr IFS_CloseDir int(10) EXTPROC('closedir');                                                        2000039
   417+   DirStructure pointer VALUE;                                                                         2000040
   418+end-pr IFS_CloseDir;                                                                                    2000041
   419+                                                                                                        2000042
```

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Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   420+                                                                                                                    2000043
   421+dcl-pr IFS_lSeek int(10) EXTPROC('lseek');                                                                          2000044
   422+   handle int(10) VALUE;                                                                                            2000045
   423+   offset int(10) VALUE;                                                                                            2000046
   424+   whence int(10) VALUE;                                                                                            2000047
   425+end-pr IFS_lSeek;                                                                                                   2000048
   426+                                                                                                                    2000049
   427+dcl-pr IFS_RenameUnlink int(10) EXTPROC('Qp0lRenameUnlink');                                                        2000050
   428+   old pointer VALUE OPTIONS(*STRING);                                                                              2000051
   429+   new pointer VALUE OPTIONS(*STRING);                                                                              2000052
   430+end-pr IFS_RenameUnlink;                                                                                            2000053
   431+                                                                                                                    2000054
   432+dcl-pr IFS_unlink int(10) EXTPROC('unlink');                                                                        2000055
   433+   FileName pointer VALUE OPTIONS(*STRING);                                                                          2000056
   434+end-pr IFS_unlink;                                                                                                  2000057
   435+                                                                                                                    2000058
   436+                                                                                                                    2000059
   437+                                                                                                                    2000060
   438+// ***************************************************************************                                       2000061
   439+//  IFS: File Access Modes                                                                                          2000062
   440+// ***************************************************************************                                       2000063
   441+                                                                                                                    2000064
   442+dcl-c #O_RDONLY 1;                                                                                                  2000065
   443+dcl-c #O_WRONLY 2;                                                                                                  2000066
   444+dcl-c #O_RDWR 4;                                                                                                    2000067
   445+dcl-c #O_CREAT 8;                                                                                                   2000068
   446+dcl-c #O_EXCL 16;                                                                                                   2000069
   447+dcl-c #O_CCSID 32;                                                                                                  2000070
   448+dcl-c #O_TRUNC 64;                                                                                                  2000071
   449+dcl-c #O_APPEND 256;                                                                                                2000072
   450+dcl-c #O_CODEPAGE 8388608;                                                                                          2000073
   451+dcl-c #O_TEXTDATA 16777216;                                                                                         2000074
   452+dcl-c #O_TEXT_CREAT 33554432;                                                                                       2000075
   453+dcl-c #O_INHERITMODE 134217728;                                                                                     2000076
   454+dcl-c #O_LARGEFILE 536870912;                                                                                       2000077
   455+                                                                                                                    2000078
   456+dcl-c #O_SYNC 1024;                                                                                                 2000079
   457+dcl-c #O_DSYNC 2048;                                                                                                2000080
   458+dcl-c #O_RSYNC 4096;                                                                                                2000081
   459+                                                                                                                    2000082
   460+dcl-c #O_SHARE_RDONLY 65536;                                                                                        2000083
   461+dcl-c #O_SHARE_WRONLY 131072;                                                                                       2000084
   462+dcl-c #O_SHARE_RDWR 262144;                                                                                         2000085
   463+dcl-c #O_SHARE_NONE 524288;                                                                                         2000086
   464+                                                                                                                    2000087
   465+dcl-c #EBUSY 3029;                                                                                                  2000088
   466+                                                                                                                    2000089
   467+// ***************************************************************************                                       2000090
   468+//  IFS: Security Attributes and File Types                                                                         2000091
   469+// ***************************************************************************                                       2000092
   470+                                                                                                                    2000093
   471+//    Authority for the file owner                                                                                  2000094
   472+                                                                                                                    2000095
   473+dcl-c #S_IRUSR 256;                                                                                                 2000096
   474+dcl-c #S_IWUSR 128;                                                                                                 2000097
   475+dcl-c #S_IXUSR 64;                                                                                                  2000098
```

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   476+dcl-c #S_IRWXU 448;                                                                                                 2000099
   477+                                                                                                                     2000100
   478+//    Authority for the primary group                                                                                2000101
   479+                                                                                                                     2000102
   480+dcl-c #S_IRGRP 32;                                                                                                   2000103
   481+dcl-c #S_IWGRP 16;                                                                                                   2000104
   482+dcl-c #S_IXGRP 8;                                                                                                    2000105
   483+dcl-c #S_IRWXG 56;                                                                                                   2000106
   484+                                                                                                                     2000107
   485+//    Authority for *public                                                                                          2000108
   486+                                                                                                                     2000109
   487+dcl-c #S_IROTH 4;                                                                                                    2000110
   488+dcl-c #S_IWOTH 2;                                                                                                    2000111
   489+dcl-c #S_IXOTH 1;                                                                                                    2000112
   490+dcl-c #S_IRWXO 7;                                                                                                    2000113
   491+dcl-c #S_IFFIFO x'0100';                                                                                             2000114
   492+dcl-c #S_IFCHR x'0200';                                                                                              2000115
   493+dcl-c #S_IFDIR x'0400';                                                                                              2000116
   494+dcl-c #S_IFBLK x'0600';                                                                                              2000117
   495+dcl-c #S_IFREG x'1000';                                                                                              2000118
   496+dcl-c #S_IFLNK x'A000';                                                                                              2000119
   497+dcl-c #S_IFSOCK x'140000';                                                                                           2000120
   498+dcl-c #S_IFNATIVE x'200000';                                                                                         2000121
   499+                                                                                                                     2000122
   500+dcl-ds IFSStatDs_T INZ TEMPLATE;                                                                                     2000123
   501+   st_mode        uns(10);                                                                                            2000124
   502+   st_ino         uns(10);                                                                                            2000125
   503+   st_nlink       uns(5);                                                                                             2000126
   504+   st_reserved2   char(2);                                                                                            2000127
   505+   st_uid         uns(10);                                                                                            2000128
   506+   st_gid         uns(10);                                                                                            2000129
   507+   st_size        uns(10);                                                                                            2000130
   508+   st_atime       int(10);                                                                                            2000131
   509+   st_mtime       int(10);                                                                                            2000132
   510+   st_ctime       int(10);                                                                                            2000133
   511+   st_dev         uns(10);                                                                                            2000134
   512+   st_blksize     uns(10);                                                                                            2000135
   513+   st_allocsize   uns(10);                                                                                            2000136
   514+   st_objtype     char(11);                                                                                           2000137
   515+   st_reserved3   char(1);                                                                                            2000138
   516+   st_codepage    uns(5);                                                                                             2000139
   517+   st_ccsid       uns(5);                                                                                             2000140
   518+   st_reserved1   char(59);                                                                                            2000141
   519+   st_ino_gen_id  uns(10);                                                                                            2000142
   520+end-ds IFSStatDs_T;                                                                                                  2000143
   521+                                                                                                                     2000144
   522+dcl-c  #_S_IFDIR x'0400';                                                                                            2000145
   523+                                                                                                                     2000146
   524+dcl-ds IFSDirEntryDs     based(PtrIFSDirEntryDs);                                                                    2000147
   525+   d_reserved1           char(16);                                                                                   2000148
   526+   d_fileno_gen_id       uns(10);                                                                                    2000149
   527+   d_fileno              int(10);                                                                                    2000150
   528+   d_reclen              uns(10);                                                                                    2000151
   529+   d_reserved3           int(10);                                                                                    2000152
   530+   d_reserved4           char(6);                                                                                    2000153
   531+   d_reserved5           char(2);                                                                                    2000154
```

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   532+   d_ccsid               int(10);                                                                                   2000155
   533+   d_country_id          char(2);                                                                                   2000156
   534+   d_language_id         char(3);                                                                                   2000157
   535+   d_reserved6           char(3);                                                                                   2000158
   536+   d_namelen             uns(10);                                                                                   2000159
   537+   d_name                char(640);                                                                                 2000160
   538+end-ds IFSDirEntryDs;                                                                                               2000161
   539+                                                                                                                    2000162
   540+dcl-c #SEEK_SET 0;                                                                                                  2000163
   541+dcl-c #SEEK_CUR 1;                                                                                                  2000164
   542+dcl-c #SEEK_END 2;                                                                                                  2000165
   543+// ---------------------------------------------------------------------------                                      2000166
   544+/INCLUDE 'QSYSCPY/DERRNO.RPGLEINC'                                                                                  2000167


Line   <---------------------- Source Specifications ---------------------------><---- Comments ----> Do  Page  Change Src Seq
Number ....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10 Num Line  Date   Id  Number
        *------------------------------------------------------------------------------------------*
        * RPG member name  . . . . . :  DERRNO.RP+                                                 *                    3
        * External name  . . . . . . :  ./QSYSCPY/DERRNO.RPGLEINC                                  *                    3
        * Last change  . . . . . . . :  26-03-27  19:38:00                                          *                    3
        *------------------------------------------------------------------------------------------*


Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   545+**FREE                                                                                                              3000001
   546+/IF DEFINED(DERRNO)                                                                                                 3000002
            LINES EXCLUDED: 1
   547+/ENDIF                                                                                                              3000004
   548+/DEFINE DERRNO                                                                                                      3000005
   549+                                                                                                                    3000006
   550+                                                                                                                    3000007
   551+Dcl-PR sys_errno pointer ExtProc('__errno');                                                                        3000008
   552+End-PR sys_errno;                                                                                                   3000009
   553+                                                                                                                    3000010
   554+Dcl-PR strerror pointer ExtProc('strerror');                                                                        3000011
   555+   errnum     int(10) value;                                                                                         3000012
   556+End-PR strerror;                                                                                                    3000013
   557+                                                                                                                    3000014
   558+Dcl-PR perror  ExtProc('perror');                                                                                   3000015
   559+   comment    pointer value options(*string);                                                                        3000016
   560+End-PR perror;                                                                                                      3000017
   561+                                                                                                                    3000018
   562+Dcl-PR errno  int(10) ExtProc(*DCLCASE);                                                                            3000019
   563+End-PR errno;                                                                                                       3000020
   564+                                                                                                                    3000021
   565+Dcl-s errnoS4 zoned(4:0);                                                                                           3000022
   566+                                                                                                                    3000023
   567+// ******************************************************************                                               3000024
   568+// * Error Numbers   (use DSPMSGD CPEnnnn to display the text)          *                                           3000025
   569+// ******************************************************************                                               3000026
   570+Dcl-ENum IFS_ERROR_NUMBERS;                                                                                         3000027
   571+                                                                                                                    3000028
   572+   #ENOENT             3025; // 3025 No path or library with this name                                              3000029
   573+   #EACCESS            3401; // 3401 Permission denied                                                                3000030
```

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   574+   #ENOTDIR            3403; // 3403 Not a directory.                                                                3000031
   575+   #ENOSPC             3404; // 3404 No space available.                                                             3000032
   576+   #EXDEV               3405; // 3405 Improper link.                                                                  3000033
   577+   #EWOULDBLOCK         3406; // 3406 Operation would have caused the process to be suspended.                        3000034
   578+   #EAGAIN              3406; // 3406                                                                                 3000035
   579+   #EINTERRUPT          3407; // 3407 Interrupted function call.                                                      3000036
   580+   #EFAULT              3408; // 3408 The address used for an argument was not correct.                               3000037
   581+   #ETIMEOUT            3409; // 3409 Operation timed out.                                                            3000038
   582+   #ENXIO               3415; // 3415 No such device or address                                                       3000039
   583+   #EADDRINUS           3420; // 3420 Address already in use.                                                         3000040
   584+   #EADDRNOTAVAIL       3421; // 3421 Address not available.                                                          3000041
   585+   #EAFNOSUPPORT        3422; // 3422 The type of socket is not supported                                             3000042
   586+   #EALREADY            3423; // 3423 Operation already in progress.                                                  3000043
   587+   #ECONNABORTED        3424; // 3424 Connection ended abnormally.                                                    3000044
   588+   #ECONNREFUSED        3425; // 3425 A remote host refused an attempted                                              3000045
   589+   #ECONNRESET          3426; // 3426 A connection with a remote socket w                                             3000046
   590+   #EDESTADDRREQ        3427; // 3427 Operation requires destination addr                                             3000047
   591+   #EHOSTDOWN           3428; // 3428 A remote host is not available.                                                 3000048
   592+   #EHOSTUNREACH        3429; // 3429 A route to the remote host is not a                                             3000049
   593+   #EINPROGRESS         3430; // 3430 Operation in progress.                                                          3000050
   594+   #EISCONN             3431; // 3431 A connection has already been estab                                             3000051
   595+   #EMSGSIZE            3432; // 3432 Message size out of range.                                                      3000052
   596+   #ENETDOWN            3433; // 3433 The network is not currently availa                                             3000053
   597+   #ENETRESET           3434; // 3434 The network is not currently availa                                             3000054
   598+   #ENETUNREACH         3435; // 3435 A socket is connected to a host tha                                             3000055
   599+   #ENOBUFS             3436; // 3436 There is not enough buffer space fo                                             3000056
   600+   #ENOPROTOOPT         3437; // 3437 The protocol does not support the s                                             3000057
   601+   #ENOTCONN            3438; // 3438 Requested operation requires a conn                                             3000058
   602+   #ENOTSOCK            3439; // 3439 The specified descriptor does not r                                             3000059
   603+   #ENOTSUP             3440; // 3440 Operation not supported.                                                        3000060
   604+   #EOPNOTSUPP          3440; // 3440 Operation not supported.                                                        3000061
   605+   #EPFNOSUPPORT        3441; // 3441 The socket protocol family is not supported.                                    3000062
   606+   #EPROTONOSUPPORT     3442; // 3442 No protocol of the specified type and domain exists.                            3000063
   607+   #EPROTOTYPE          3443; // 3443 The socket type or protocols are not compatible.                                3000064
   608+   #ERCVDERR            3444; // 3444 An error indication was sent by the                                             3000065
   609+   #ESHUTDOWN           3445; // 3445 Cannot send data after a shutdown.                                              3000066
   610+   #ESOCKTNOSUPPORT     3446; // 3446 The specified socket type is not su                                             3000067
   611+   #ETIMEDOUT           3447; // 3447 A remote host did not respond within a reasonable time.                         3000068
   612+   #EUNATCH             3448; // 3448 The protocol required to support the specified address                          3000069
   613+   //                                 family is not available at this time.                                             3000070
   614+   #EBADF               3450; // 3450 Descriptor not valid.                                                           3000071
   615+   #EMFILE              3452; // 3452 Too many open files for this proces                                             3000072
   616+   #ENFILE              3453; // 3453 Too many open files in the system.                                              3000073
   617+   #EPIPE               3455; // 3455 Broken pipe.                                                                    3000074
   618+   #EEXIST              3457; // 3457 File exists.                                                                    3000075
   619+   #EDEADLK             3459; // 3459 Resource deadlock avoided.                                                      3000076
   620+   #ENOMEM              3460; // 3460 Storage allocation request failed.                                              3000077
   621+   #EOWNERTERM          3462; // 3462 The mutex no longer exists because                                              3000078
   622+   #EDESTROYED          3463; // 3463 The mutex was destroyed, or the obj                                             3000079
   623+   #ETERM               3464; // 3464 Operation terminated.                                                           3000080
   624+   #EMLINK              3468; // 3468 Maximum link count for a file was e                                             3000081
   625+   #ESPIPE              3469; // 3469 Seek request not supported for obje                                             3000082
   626+   #ENOSYS              3470; // 3470 Function not implemented.                                                       3000083
   627+   #EISDIR              3471; // 3471 Specified target is a directory.                                                3000084
   628+   #EROFS               3472; // 3472 Read-only file system.                                                          3000085
   629+   #EUNKNOWN            3474; // 3474 Unknown system state.                                                           3000086
```

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Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   630+   #EITERBAD           3475; // 3475 Iterator is invalid.                                                   3000087
   631+   #EDAMAGE            3484; // 3484 A damaged object was encountered.                                      3000088
   632+   #ELOOP              3485; // 3485 A loop exists in the symbolic links.                                   3000089
   633+   #ENAMETOOLONG       3486; // 3486 A path name is too long.                                               3000090
   634+   #ENOLCK             3487; // 3487 No locks available                                                     3000091
   635+   #ENOTEMPTY          3488; // 3488 Directory not empty.                                                   3000092
   636+   #ENOSYSRSC          3489; // 3489 System resources not available to c                                    3000093
   637+   #ECONVERT           3490; // 3490 Conversion error.                                                      3000094
   638+   #E2BIG              3491; // 3491 Argument list too long.                                                3000095
   639+   #EILSEQ             3492; // 3492 Conversion stopped due to input cha                                    3000096
   640+   #ENOTENROLL         3498; // 3498 User not enrolled in system distrib                                    3000097
   641+   #EOFFLINE           3499; // 3499 Object is suspended.                                                   3000098
   642+   #EROOBJ             3500; // 3500 Object is a read only object.                                          3000099
   643+   #ELOCKED            3506; // 3506 Area being read from or written to                                     3000100
   644+   #EFBIG              3507; // 3507 Object too large.                                                      3000101
   645+   #EIDRM              3509; // 3509 The semaphore, shared memory, or me                                    3000102
   646+   #ENOMSG             3510; // 3510 The queue does not contain a messag                                    3000103
   647+   #EFILECVT           3511; // 3511 File ID conversion of a directory f                                    3000104
   648+   #EBADFID            3512; // 3512 A File ID could not be assigned whe                                    3000105
   649+   #ESTALE             3513; // 3513 A File ID could not be assigned whe                                    3000106
   650+   #ESRCH              3515; // 3515 No such process.                                                       3000107
   651+   #ENOTSIGINIT        3516; // 3516 Process not enabled for signals.                                       3000108
   652+   #ECHILD             3517; // 3517 No child process.                                                      3000109
   653+   #ETOOMANYREFS       3523; // 3523 Too many references.                                                   3000110
   654+   #ENOTSAFE           3524; // 3524 Function not allowed.                                                  3000111
   655+                                                                                                            3000112
   656+End-ENum;                                                                                                   3000113
   657 /INCLUDE 'QSYSCPY/PERRNO.RPGLEINC'                                                                           000106


Line   <---------------------- Source Specifications ---------------------------><---- Comments ----> Do  Page  Change Src Seq
Number ....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10 Num Line  Date   Id  Number
        *-----------------------------------------------------------------------------------------*
        * RPG member name  . . . . . :  PERRNO.RP+                                                *                     4
        * External name  . . . . . . :  ./QSYSCPY/PERRNO.RPGLEINC                                  *                     4
        * Last change  . . . . . . . :  26-03-26  09:18:32                                         *                     4
        *-----------------------------------------------------------------------------------------*
   658+P errno           B                                                                                            4000001
   659+D errno           PI            10I 0                                                                          4000002
   660+D p_errno         S               *                                                                            4000003
   661+D wwreturn        S             10I 0 based(p_errno)                                                           4000004
   662+C                   eval      p_errno = sys_errno                                                              4000005
   663+c                   return    wwreturn                                                                         4000006
   664+P                 E                                                                                            4000007


Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   665                                                                                                              000107
   666                                                                                                              000108
   667                                                                                                              000109
   668 // ==========================================================================                                000110
   669 // M A I N                                                                                                   000111
   670 // ==========================================================================                                000112
   671 Dcl-Proc MAIN;                                                                                               000113
   672    Dcl-PI *N;                                                                                                000114
```

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   673       p_SplfName        CHAR( 10) CONST;                                                                          000115
   674       p_QualJob         CHAR( 26) CONST;                                                                          000116
   675       p_SplfNbr          INT( 10) CONST;                                                                          000117
   676       p_PDF_Path     VARCHAR(128) CONST;                                                                          000118
   677       p_PDF_File     VARCHAR(128) CONST;                                                                          000119
   678       p_PaperFormat  VARCHAR( 20) CONST;                                                                          000120
   679       p_PageRotation    CHAR( 10) CONST;                                                                          000121
   680       p_Margins       LIKEDS(RPDFG_Margins_T)  CONST;                                                             000122
   681       p_TimeZones     LIKEDS(RPDFG_TimeZone_T) CONST;                                                             000123
   682       p_Options       LIKEDS(RPDFG_Options_T)  CONST;                                                             000124
   683    End-PI;                                                                                                        000125
   684                                                                                                                   000126
   685    Dcl-DS h_QualJob QUALIFIED;                                                                                    000127
   686       Name        CHAR(10);                                                                                       000128
   687       User        CHAR(10);                                                                                       000129
   688       Number      CHAR( 6);                                                                                       000130
   689    End-DS;                                                                                                        000131
   690    Dcl-S  h_Job            VARCHAR(28);                                                                           000132
   691    Dcl-S  h_SplfName       CHAR(10);                                                                              000133
   692    Dcl-S  h_SplfNbr        INT(10);                                                                               000134
   693    Dcl-S  h_SplfNbrChar    CHAR(5);                                                                               000135
   694                                                                                                                   000136
   695    Dcl-S  TotalPageCount   INT(10) Inz(0);                                                                        000137
   696    Dcl-S  PageCount        INT(10) Inz(0);                                                                        000138
   697                                                                                                                   000139
   698    Dcl-S  Offset           INT(10) Inz(0);                                                                        000140
   699                                                                                                                   000141
   700    Dcl-S  h_OutBuffer      VARCHAR(100000) INZ('') CCSID(*UTF8);                                                  000142
   701                                                                                                                   000143
   702    Dcl-S  h_Cmd            VARCHAR(2000) INZ('');                                                                 000144
   703                                                                                                                   000145
   704    // various object indices                                                                                      000146
   705    Dcl-S  i_Root           INT(10);                                                                               000147
   706    Dcl-S  i_Fn             INT(10);                                                                               000148
   707    Dcl-S  i_Font           INT(10);                                                                               000149
   708    Dcl-S  h_Font           INT(10);                                                                               000150
   709    Dcl-S  i_PagesRoot      INT(10);                                                                               000151
   710    Dcl-S  i_Pages          INT(10) INZ(-1);                                                                       000152
   711    Dcl-S  i_Contents       INT(10) INZ(-1);                                                                       000153
   712    Dcl-S  i_Page           INT(10);                                                                               000154
   713    Dcl-S  i_Info           INT(10);                                                                               000155
   714    Dcl-S  i_Xref           INT(10);                                                                               000156
   715    Dcl-S  i_Trailer        INT(10);                                                                               000157
   716                                                                                                                   000158
   717    Dcl-S  pdfObjRef_Root   LIKE(RPDFG_ref);                                                                       000159
   718    Dcl-S  pdfObjRef_Font1  LIKE(RPDFG_ref);                                                                       000160
   719    Dcl-S  pdfObjRef_Font2  LIKE(RPDFG_ref);                                                                       000161
   720    Dcl-S  pdfObjRef_Info   LIKE(RPDFG_ref);                                                                       000162
   721                                                                                                                   000163
   722    Dcl-S  h_startxref      INT(10) INZ(0);                                                                        000164
   723                                                                                                                   000165
   724    // pages depth - needed for page tree structure                                                                000166
   725    // how many elements are on each level of the page tree                                                        000167
   726    Dcl-DS h_PageDepth      DIM(10) QUALIFIED;                                                                     000168
   727       count               INT(10) INZ(0);                                                                         000169
   728       objIdx              INT(10) INZ(0);                                                                         000170
```

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   729    End-DS;                                                                                                          000171
   730    Dcl-S  h_PageDepthMax   INT(10);                                                                                 000172
   731    // how many Page-objects fit into all Pages objects                                                              000173
   732    Dcl-S  h_DepthMaxPages  INT(10);                                                                                 000174
   733                                                                                                                     000175
   734    Dcl-S  h_Resources_string1 VARCHAR(1000) CCSID(*UTF8);                                                           000176
   735    Dcl-S  h_Resources_string2 VARCHAR(1000) CCSID(*UTF8);                                                           000177
   736    Dcl-S  h_MediaBox_string   LIKE(RPDFG_getMediaBoxString);                                                        000178
   737                                                                                                                     000179
   738    Dcl-S  h_streamOptions     VARCHAR(8000:4) CCSID(*HEX);                                                          000180
   739                                                                                                                     000181
   740    Dcl-DS SplfAtr LIKEDS(RPDFG_SplfAtr_T) DIM(*AUTO:2);                                                             000182
   741                                                                                                                     000183
   742    Dcl-DS SplDta QUALIFIED;                                                                                         000184
   743       FCFC        CHAR(  1);                                                                                        000185
   744       LINE        CHAR(378);                                                                                        000186
   745    End-DS;                                                                                                          000187
   746                                                                                                                     000188
   747    Dcl-ENum FCFC QUALIFIED;                                                                                         000189
   748       SPACE      ' ';                                                                                               000190
   749       SPACE2     '0';                                                                                               000191
   750       SPACE3     '-';                                                                                               000192
   751       SUPPRESS   '+';                                                                                               000193
   752       CHANNEL01  '1';                                                                                               000194
   753       CHANNEL02  '2';                                                                                               000195
   754       CHANNEL03  '3';                                                                                               000196
   755       CHANNEL04  '4';                                                                                               000197
   756       CHANNEL05  '5';                                                                                               000198
   757       CHANNEL06  '6';                                                                                               000199
   758       CHANNEL07  '7';                                                                                               000200
   759       CHANNEL08  '8';                                                                                               000201
   760       CHANNEL09  '9';                                                                                               000202
   761       CHANNEL10  'A';                                                                                               000203
   762       CHANNEL11  'B';                                                                                               000204
   763       CHANNEL12  'C';                                                                                               000205
   764    End-ENum;                                                                                                        000206
   765                                                                                                                     000207
   766    Dcl-S  h_LineNumber INT(10);                                                                                     000208
   767    Dcl-S  h_PosX       PACKED(10:4);                                                                                000209
   768    Dcl-S  h_PosY       PACKED(10:4);                                                                                000210
   769                                                                                                                     000211
   770    Dcl-S  #i           INT(10);                                                                                     000212
   771    Dcl-S  #d           INT(10);                                                                                     000213
   772                                                                                                                     000214
   773    Dcl-S  Sav_SQLCOD   LIKE(SQLCOD);                                                                                000215
   774                                                                                                                     000216
   775    // For output - bad idea                                                                                         000217
   776    Dcl-DS xref QUALIFIED;                                                                                           000218
   777       ByteOffset    ZONED(10);                                                                                       000219
   778       ObjNumber       INT(10);                                                                                       000220
   779       GenNumber       INT(10);                                                                                       000221
   780       InUse           CHAR(1);                                                                                       000222
   781    End-DS;                                                                                                          000223
   782                                                                                                                     000224
   783    // message data for message id RPDF0001                                                                          000225
   784    Dcl-DS RPDF0001 QUALIFIED;                                                                                       000226
```

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Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   785       SplfName   CHAR(10);                                                                                          000227
   786       JobName    CHAR(10);                                                                                          000228
   787       JobUser    CHAR(10);                                                                                          000229
   788       JobNumber  CHAR(6);                                                                                           000230
   789       SplfNbr    CHAR(5);                                                                                           000231
   790    End-DS;                                                                                                          000232
   791    // ---------------------------------------------------------------------                                         000233
   792                                                                                                                     000234
   793                                                                                                                     000235
   794                                                                                                                     000236
   795    // ---------------------------------------------------------------------                                         000237
   796    RPDFG_initPaperFormat(p_PaperFormat:                                                                             000238
   797                          p_Margins.Left:                                                                            000239
   798                          p_Margins.Top:                                                                             000240
   799                          p_Margins.Right:                                                                           000241
   800                          p_Margins.Bottom);                                                                         000242
   801    // ---------------------------------------------------------------------                                         000243
   802    h_QualJob = p_QualJob;                                                                                           000244
   803    IF h_QualJob.Name = '*';                                                                                         000245
   804       h_QualJob.Name   = PSDS.JobName;                                                                              000246
   805       h_QualJob.User   = PSDS.JobUser;                                                                              000247
   806       h_QualJob.Number = PSDS.JobNumber;                                                                            000248
   807    ENDIF;                                                                                                           000249
   808    h_Job = %TRIMR(h_QualJob.Number) + '/' +                                                                         000250
   809            %TRIMR(h_QualJob.User) + '/' +                                                                           000251
   810            %TRIMR(h_QualJob.Name);                                                                                  000252
   811    h_SplfName = p_SplfName;                                                                                         000253
   812    h_SplfNbr  = p_SplfNbr;                                                                                          000254
   813                                                                                                                     000255
   814                                                                                                                     000256
   815    // retrieve the spooled file attributes and check that we got only one spooled file                              000257
   816    EXEC SQL DECLARE C1 CURSOR FOR                                                                                   000258
   817             SELECT OUTPUT_QUEUE_NAME,                                                                               000259
   818                    OUTPUT_QUEUE_LIBRARY_NAME,                                                                       000260
   819                    SPOOLED_FILE_NAME,                                                                               000261
   820                    FILE_NUMBER,                                                                                     000262
   821                    USER_NAME,                                                                                       000263
   822                    STRIP(COALESCE(USER_DATA, 'no_usrdta')) USER_DATA,                                               000264
   823                    JOB_NAME,                                                                                        000265
   824                    CREATE_TIMESTAMP,                                                                                000266
   825                    TOTAL_PAGES,                                                                                     000267
   826                    STRIP(FORM_TYPE),                                                                                000268
   827                    PAGE_LENGTH,                                                                                     000269
   828                    PAGE_WIDTH,                                                                                      000270
   829                    PAGE_ROTATION                                                                                    000271
   830              FROM QSYS2.OUTPUT_QUEUE_ENTRIES                                                                        000272
   831             WHERE SPOOLED_FILE_NAME = :h_SplfName                                                                   000273
   832               AND JOB_NAME          = :h_Job                                                                        000274
   833               AND (FILE_NUMBER      = :h_SplfNbr OR :h_SplfNbr < 1)                                                 000275
   834             ORDER BY FILE_NUMBER DESC                                                                               000276
   835             LIMIT 2;                                                                                                000277
   836                                                                                                                     000278
   837    EXEC SQL OPEN C1;                                                                                                000279
   838    Sav_SQLCOD = SQLCOD;                                                                                             000280
   839    IF Sav_SQLCOD = SQL_OK;                                                                                          000281
   840       EXEC SQL FETCH C1                                                                                             000282
```

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   841                 FOR 2 ROWS                                                                                          000283
   842                INTO :SplfAtr;                                                                                       000284
   843       Sav_SQLCOD = SQLCOD;                                                                                          000285
   844    ENDIF;                                                                                                           000286
   845    EXEC SQL CLOSE C1;                                                                                               000287
   846                                                                                                                     000288
   847    SELECT;                                                                                                          000289
   848       WHEN Sav_SQLCOD = SQL_OK OR                                                                                   000290
   849            (Sav_SQLCOD > 0 AND                                                                                      000291
   850             Sav_SQLCOD <> SQL_ERROR_NOTFOUND);                                                                      000292
   851         IF p_SplfNbr = SPLFNBR_ONLY_NUM AND                                                                         000293
   852            %ELEM(SplfAtr) > 1;                                                                                      000294
   853            // TODO: nicht eindeutig, mehrere Spooldateien gefunden                                                  000295
   854         ENDIF;                                                                                                      000296
   855       WHEN Sav_SQLCOD = SQL_ERROR_NOTFOUND;                                                                         000297
   856         RPDF0001.SplFName = h_SplfName;                                                                             000298
   857         RPDF0001.JobName  = h_QualJob.Name;                                                                         000299
   858         RPDF0001.JobUser  = h_QualJob.User;                                                                         000300
   859         RPDF0001.JobNumber= h_QualJob.Number;                                                                       000301
   860         SELECT;                                                                                                     000302
   861            WHEN p_SplfNbr = SPLFNBR_ONLY_NUM;                                                                       000303
   862               h_SplfNbrChar = '*ONLY';                                                                              000304
   863            WHEN p_SplfNbr = SPLFNBR_LAST_NUM;                                                                       000305
   864               h_SplfNbrChar = '*LAST';                                                                              000306
   865            OTHER;                                                                                                   000307
   866               h_SplfNbrChar = %CHAR(p_SplfNbr);                                                                     000308
   867         ENDSL;                                                                                                      000309
   868         RPDF0001.SplfNbr  = h_SplfNbrChar;                                                                          000310
   869         SND-MSG *ESCAPE %MSG(MSGID.RPDF001:RPDFGMSGF:RPDF0001)                                                      000311
   870                         %TARGET(*CTLBDY:1);                                                                         000312
   871       OTHER;                                                                                                        000313
   872         // TODO: SQL error handling                                                                                 000314
   873         // SND-MSG *ESCAPE %MSG('RPDF001':RPDFGMSGF:RPDF0001)                                                       000315
   874         //                 %TARGET(*CTLBDY:1);                                                                      000316
   875    ENDSL;                                                                                                           000317
   876                                                                                                                     000318
   877    RPDFG_openPdfFile(p_PDF_Path:p_PDF_File:SplfAtr(1));                                                             000319
   878                                                                                                                     000320
   879    MONITOR;                                                                                                         000321
   880       h_Cmd = 'CRTPF FILE(QTEMP/RPDFGCVT) RCDLEN(387) SIZE(*NOMAX)';                                                000322
   881       QCMDEXC(h_Cmd:%LEN(h_Cmd));                                                                                   000323
   882    ON-ERROR;                                                                                                        000324
   883    ENDMON;                                                                                                          000325
   884                                                                                                                     000326
   885    h_Cmd = 'CPYSPLF FILE(' + SplfAtr(1).SplfName +')' +                                                             000327
   886                   ' TOFILE(QTEMP/RPDFGCVT)' +                                                                       000328
   887                   ' JOB(' + h_Job + ')' +                                                                           000329
   888                   ' SPLNBR(' + %CHAR(h_SplfNbr) + ')' +                                                             000330
   889                   ' CTLCHAR(*FCFC)';                                                                                000331
   890    QCMDEXC(h_Cmd:%LEN(h_Cmd));                                                                                      000332
   891                                                                                                                     000333
   892    EXEC SQL DECLARE C2 CURSOR FOR                                                                                   000334
   893             SELECT SUBSTR(RPDFGCVT, 1, 1) FCFC,                                                                     000335
   894                    SUBSTR(RPDFGCVT, 2, 378) LINE                                                                    000336
   895               FROM QTEMP.RPDFGCVT;                                                                                  000337
   896                                                                                                                     000338
```

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   897    EXEC SQL OPEN C2;                                                                                                 000339
   898                                                                                                                      000340
   899    // ==========================================================================                                     000341
   900    // --------------------------------------------------------------------------                                     000342
   901    // actual PDF creation                                                                                            000343
   902    RPDFG_calculatePrintingValues(p_PageRotation:                                                                     000344
   903                                  SplfAtr(1).Page_Rotation:                                                            000345
   904                                  SplfAtr(1).Page_Length:                                                              000346
   905                                  SplfAtr(1).Page_Width);                                                              000347
   906                                                                                                                      000348
   907    // Strategy for large PDF files with many pages:                                                                  000349
   908    // The PDF-file has to know it's pages-page-tree with parent kids and count                                       000350
   909    // of kids+grandchildren.                                                                                         000351
   910    // I can't hold all that in the storage at the same time.                                                         000352
   911    // So some mechanism is necessary to minimize storage consumption.                                               000353
   912    // root --> root-Pages [kids, count of ALL kids in this branch]                                                   000354
   913    //           --> Pages [Parent, kids, count]                                                                      000355
   914    // ------------------------------------------------------------------                                             000356
   915    // Calculate the maximum depth of the page tree with the number of pages                                          000357
   916    // and the number of kids per Pages object.                                                                       000358
   917    h_PageDepthMax  = 0;                                                                                              000359
   918    h_DepthMaxPages = 1;                                                                                              000360
   919    DOU h_DepthMaxPages > SplfAtr(1).Total_Pages;                                                                     000361
   920       h_PageDepthMax  += 1;                                                                                          000362
   921       h_DepthMaxPages  = h_DepthMaxPages * PAGE_PER_PAGES;                                                           000363
   922    ENDDO;                                                                                                            000364
   923                                                                                                                      000365
   924                                                                                                                      000366
   925                                                                                                                      000367
   926    // root object                                                                                                    000368
   927    i_Root    = RPDFG_newObject();                                                                                    000369
   928    RPDFG_addDictEntry(i_Root:'Type':'/Catalog');                                                                     000370
   929    pdfObjRef_Root = RPDFG_ref(i_Root);                                                                               000371
   930                                                                                                                      000372
   931    // Pages object                                                                                                   000373
   932    i_Pages   = RPDFG_newObject();                                                                                    000374
   933    RPDFG_addDictEntry(i_Pages:'Type':'/Pages');                                                                      000375
   934    // put this in the tree                                                                                           000376
   935    h_PageDepth(1).count  = 1;                                                                                        000377
   936    h_PageDepth(1).objIdx = i_Pages;                                                                                  000378
   937    // add the reference to the Pages object to the root object                                                       000379
   938    RPDFG_addDictEntry(i_Root:'Pages' :RPDFG_ref(i_Pages));                                                           000380
   939                                                                                                                      000381
   940    // "Root" is now ready and can be written; Object Index will be re-used for antoher object                        000382
   941    RPDFG_serializeObject(i_Root);                                                                                    000383
   942                                                                                                                      000384
   943                                                                                                                      000385
   944    // Page - Font - F1 dictionary                                                                                    000386
   945    i_Fn = RPDFG_newObject(RPDFG_SPECIAL_OBJECT.INTERNAL);                                                            000387
   946                                                                                                                      000388
   947    // Font object 1 - Courier, used for the whole document                                                           000389
   948    i_Font    = RPDFG_newObject();                                                                                    000390
   949    RPDFG_addDictEntry(i_Font:'Type'    :'/Font');                                                                    000391
   950    RPDFG_addDictEntry(i_Font:'Subtype' :'/Type1');                                                                   000392
   951    RPDFG_addDictEntry(i_Font:'BaseFont':'/Courier');                                                                 000393
   952    RPDFG_addDictEntry(i_Font:'Encoding':'/WinAnsiEncoding');                                                         000394
```

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```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
   953    // remember the reference to the font object in the PDF file                                                    000395
   954    pdfObjRef_Font1 = RPDFG_ref(i_Font);                                                                            000396
   955    // Font is ready                                                                                                000397
   956    RPDFG_serializeObject(i_Font);                                                                                  000398
   957    RPDFG_addDictEntry(i_Fn:'F1':pdfObjRef_Font1);                                                                  000399
   958                                                                                                                    000400
   959    // Extras need a second font                                                                                    000401
   960    IF OPTIONS.NONE IN p_Options.value;                                                                             000402
   961       // no optional extras                                                                                        000403
   962    ELSE;                                                                                                           000404
   963       // Font object 2 - used for the line numbers when option *NUMBERS is specified                               000405
   964       i_Font    = RPDFG_newObject();                                                                              000406
   965       RPDFG_addDictEntry(i_Font:'Type'    :'/Font');                                                              000407
   966       RPDFG_addDictEntry(i_Font:'Subtype' :'/Type1');                                                             000408
   967       RPDFG_addDictEntry(i_Font:'BaseFont':'/Helvetica');                                                         000409
   968       RPDFG_addDictEntry(i_Font:'Encoding':'/WinAnsiEncoding');                                                   000410
   969       // remember the reference to the font object in the PDF file                                                000411
   970       pdfObjRef_Font2 = RPDFG_ref(i_Font);                                                                        000412
   971       // Font is ready                                                                                            000413
   972       RPDFG_serializeObject(i_Font);                                                                              000414
   973       RPDFG_addDictEntry(i_Fn:'F2':pdfObjRef_Font2);                                                              000415
   974    ENDIF;                                                                                                          000416
   975    // -----------------------------------------------------                                                        000417
   976    // create reusable h_Resources_string for the Page object                                                       000418
   977    // -----------------------------------------------------                                                        000419
   978    // Page - Font dictionary                                                                                       000420
   979    h_Font = RPDFG_newObject(RPDFG_SPECIAL_OBJECT.INTERNAL);                                                        000421
   980    RPDFG_addDictEntry(h_Font:'Font':RPDFG_serializeDict(RPDFG_getObjDictIdx(i_Fn)));                               000422
   981    h_Resources_string1 = RPDFG_serializeDict(RPDFG_getObjDictIdx(h_Font));                                         000423
   982    // The dictionaries of Fn and h_Font are now gone, so the objects have to be deleted, too!!!                    000424
   983    RPDFG_deleteObject(i_Fn);                                                                                       000425
   984    RPDFG_deleteObject(h_Font);                                                                                     000426
   985                                                                                                                    000427
   986    %LEN(h_streamOptions) = 0;                                                                                      000428
   987    IF OPTIONS.NONE IN p_Options.value;                                                                             000429
   988       // no optional extras                                                                                        000430
   989    ELSE;                                                                                                           000431
   990       // Page - Font - F2 dictionary                                                                               000432
   991       i_Fn = RPDFG_newObject(RPDFG_SPECIAL_OBJECT.INTERNAL);                                                      000433
   992       RPDFG_addDictEntry(i_Fn:'F2':pdfObjRef_Font2);                                                              000434
   993       // Page - Font dictionary                                                                                   000435
   994       h_Font = RPDFG_newObject(RPDFG_SPECIAL_OBJECT.INTERNAL);                                                    000436
   995       RPDFG_addDictEntry(h_Font:'Font':RPDFG_serializeDict(RPDFG_getObjDictIdx(i_Fn)));                           000437
   996       h_Resources_string2 = RPDFG_serializeDict(RPDFG_getObjDictIdx(h_Font));                                     000438
   997       // The dictionaries of Fn and h_Font are now gone, so the objects have to be deleted, too!!!                 000439
   998       RPDFG_deleteObject(i_Fn);                                                                                   000440
   999       RPDFG_deleteObject(h_Font);                                                                                 000441
  1000       // ------------------------------------------------------------------------                                 000442
  1001       %LEN(h_streamOptions) = 0;                                                                                  000443
  1002       IF OPTIONS.JOBINFO IN p_Options.value;                                                                      000444
  1003          RPDFG_createPrePrintForm_JobInfo(%ADDR(h_streamOptions));                                                000445
  1004       ENDIF;                                                                                                      000446
  1005       IF OPTIONS.NUMBERS IN p_Options.value;                                                                      000447
  1006          RPDFG_createPrePrintForm_Numbers(%ADDR(h_streamOptions));                                                000448
  1007       ENDIF;                                                                                                      000449
  1008    ENDIF;                                                                                                          000450
```

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
  1009    // ----------------------------------------------------------------------                                  000451
  1010    // -------------------------------------------------------                                                 000452
  1011    // This one is needed for every page, so we create it once and reuse the string                            000453
  1012    h_MediaBox_string = RPDFG_getMediaBoxString();                                                             000454
  1013    // ---------------------------------------------------------------------------                             000455
  1014    // check the tree depth and create new "Pages" objects                                                     000456
  1015    FOR #d = 2 TO h_PageDepthMax - 1;                                                                          000457
  1016       IF h_PageDepth(#d).count < 1;                                                                           000458
  1017          h_PageDepth(#d).count += 1;                                                                          000459
  1018          h_PageDepth(#d).objIdx = RPDFG_newObject();                                                          000460
  1019          RPDFG_addDictEntry(h_PageDepth(#d).objIdx:'Type':'/Pages');                                          000461
  1020          // connect the new "Pages" object to it's parent in the tree                                         000462
  1021          RPDFG_addDictEntry(h_PageDepth(#d).objIdx:'Parent' :RPDFG_ref(h_PageDepth(#d-1).objIdx));            000463
  1022       ENDIF;                                                                                                  000464
  1023    ENDFOR;                                                                                                    000465
  1024    // ---------------------------------------------------------------------                                   000466
  1025                                                                                                               000467
  1026                                                                                                               000468
  1027    // = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = =                  000469
  1028    DOU 'all' = 'data read';                                                                                   000470
  1029                                                                                                               000471
  1030       EXEC SQL FETCH C2 INTO :SplDta;                                                                         000472
  1031       IF SQLCOD <> SQL_OK;                                                                                    000473
  1032          LEAVE;                                                                                               000474
  1033       ENDIF;                                                                                                  000475
  1034                                                                                                               000476
  1035       IF SplDta.FCFC = FCFC.CHANNEL01;                                                                        000477
  1036                                                                                                               000478
  1037          IF PageCount > 0;                                                                                    000479
  1038             RPDFG_serializeObject(i_Page);                                                                    000480
  1039             RPDFG_serializeObject(i_Contents);                                                                000481
  1040          ENDIF;                                                                                               000482
  1041                                                                                                               000483
  1042          PageCount += 1;                                                                                      000484
  1043                                                                                                               000485
  1044          // check if the current "Pages" tree has place for another "Page" object,                            000486
  1045          // otherwise create a new "Pages" object and so on                                                   000487
  1046          FOR #d = 2 TO h_PageDepthMax - 1;                                                                    000488
  1047             IF RPDFG_getNumberOfKids(h_PageDepth(#d).objIdx) = PAGES_PER_PAGES;                               000489
  1048                // full, write it                                                                              000490
  1049                RPDFG_serializeObject(h_PageDepth(#d).objIdx);                                                 000491
  1050                // we need a new one                                                                           000492
  1051                h_PageDepth(#d).objIdx = RPDFG_newObject();                                                    000493
  1052                RPDFG_setRelation(h_PageDepth(#d-1).objIdx:                                                    000494
  1053                                  h_PageDepth(#d).objIdx);                                                     000495
  1054             ENDIF;                                                                                            000496
  1055          ENDFOR;                                                                                              000497
  1056                                                                                                               000498
  1057          IF h_PageDepthMax > 1;                                                                               000499
  1058             // the current Pages object is the one on the max depth level - 1                                 000500
  1059             i_Pages = h_PageDepth(h_PageDepthMax-1).objIdx;                                                   000501
  1060          ELSE;                                                                                                000502
  1061             // There is only 1 level when the number of total pages < PAGE_PER_PAGES                          000503
  1062             i_Pages = h_PageDepth(1).objIdx;                                                                  000504
  1063          ENDIF;                                                                                               000505
  1064          // but is there place for another "Page" object?                                                     000506
```

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
  1065         IF RPDFG_getNumberOfKids(i_Pages) = PAGE_PER_PAGES;                                                  000507
  1066            // full, write it                                                                                  000508
  1067            RPDFG_serializeObject(i_Pages);                                                                   000509
  1068            // we need a new one                                                                               000510
  1069            i_Pages = RPDFG_newObject();                                                                      000511
  1070            h_PageDepth(#d).objIdx = i_Pages;                                                                 000512
  1071         ENDIF;                                                                                               000513
  1072                                                                                                              000514
  1073         // create Page object                                                                                000515
  1074         i_Page = RPDFG_newObject();                                                                          000516
  1075         RPDFG_addDictEntry(i_Page:'Type':'/Page');                                                           000517
  1076         // add the Page object ref to the current Pages object                                               000518
  1077         // RPDFG_addObjKidRef(i_Pages:i_Page);                                                               000519
  1078         // RPDFG_g_obj(i_Pages).NbrOfKids += 1;                                                              000520
  1079         // RPDFG_g_obj(i_Pages).Kids(RPDFG_g_obj(i_Pages).NbrOfKids) = i_Page;                               000521
  1080         // // the parent of the "Page" is the current "Pages" object                                         000522
  1081         // RPDFG_addDictEntry(i_Page:'Parent':RPDFG_ref(i_Pages));                                           000523
  1082         RPDFG_setRelation(i_Pages:i_Page);                                                                   000524
  1083         RPDFG_addDictEntry(i_Page:'Resources':h_Resources_string1);                                          000525
  1084         RPDFG_addDictEntry(i_Page:'MediaBox':h_MediaBox_string);                                             000526
  1085         // RPDFG_addDictEntry(i_Page:'Rotate':%CHAR(RPDFG_getRotate()));                                     000527
  1086                                                                                                              000528
  1087         h_LineNumber = 0;                                                                                    000529
  1088                                                                                                              000530
  1089         i_Contents = RPDFG_newObject(RPDFG_SPECIAL_OBJECT.CONTENTS);                                         000531
  1090         RPDFG_addDictEntry(i_Page:'Contents':RPDFG_ref(i_Contents));                                         000532
  1091                                                                                                              000533
  1092         IF %LEN(h_streamOptions) > 0;                                                                        000534
  1093            RPDFG_startStream();                                                                              000535
  1094            RPDFG_addToStreamHex(%ADDR(h_streamOptions:*DATA):%LEN(h_streamOptions));                         000536
  1095         ENDIF;                                                                                               000537
  1096                                                                                                              000538
  1097      ENDIF; // new page                                                                                      000539
  1098      // ------------------------------------------------                                                     000540
  1099                                                                                                              000541
  1100                                                                                                              000542
  1101      // RPDFG_calcLinePos(h_LineNumber:1:h_PosX:h_PosY);                                                     000543
  1102                                                                                                              000544
  1103      // RPDFG_addToStream(PDF_TEXTBLOCK_BEGIN + CCSID_HEX.EOL +                                              000545
  1104      //          '/F1 ' + %CHAR(g_PDF.FontSize) + PDF_TEXT_FONT_RESSOURCE + CCSID_HEX.EOL +                  000546
  1105      //          %CHAR(h_PosX) + ' ' + %CHAR(h_PosY) + ' Td' + CCSID_HEX.EOL +                               000547
  1106      //          '(' + %TRIMR(SplDta.LINE) + ') Tj' + CCSID_HEX.EOL +                                        000548
  1107      //          PDF_TEXTBLOCK_END + CCSID_HEX.EOL);                                                         000549
  1108                                                                                                              000550
  1109      SELECT;                                                                                                 000551
  1110         WHEN SplDta.FCFC = FCFC.SUPPRESS;                                                                    000552
  1111            // do nothing, stay on the same line                                                              000553
  1112         WHEN SplDta.FCFC = FCFC.SPACE2;                                                                      000554
  1113            h_LineNumber += 2;                                                                                000555
  1114         WHEN SplDta.FCFC = FCFC.SPACE3;                                                                      000556
  1115            h_LineNumber += 3;                                                                                000557
  1116         OTHER;                                                                                               000558
  1117            h_LineNumber += 1;                                                                                000559
  1118      ENDSL;                                                                                                  000560
  1119                                                                                                              000561
  1120      // we do not need to write empty lines                                                                  000562
```

-=* http://pub400.com *=-

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
  1121      IF %TRIMR(SplDta.LINE) <> '';                                                                                 000563
  1122         RPDFG_addToStreamTextBlock1252(h_LineNumber:                                                               000564
  1123                                         1:                                                                          000565
  1124                                         SplDta.LINE);                                                               000566
  1125         // RPDFG_addToStreamTextBlock(h_LineNumber:                                                                000567
  1126         //                            1:                                                                            000568
  1127         //                            SplDta.LINE);                                                                 000569
  1128         // no idea why this won't work                                                                             000570
  1129         // RPDFG_addToStreamTextBlockUcs2(h_LineNumber:                                                            000571
  1130         //                                1:                                                                          000572
  1131         //                                SplDta.LINE);                                                               000573
  1132      ENDIF;                                                                                                         000574
  1133                                                                                                                     000575
  1134                                                                                                                     000576
  1135   ENDDO; // read all spooled file lines                                                                             000577
  1136                                                                                                                     000578
  1137   // = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = = 000579
  1138                                                                                                                     000580
  1139   EXEC SQL CLOSE C2;                                                                                                000581
  1140                                                                                                                     000582
  1141   RPDFG_serializeObject(i_Page);                                                                                    000583
  1142   RPDFG_serializeObject(i_Contents);                                                                                000584
  1143                                                                                                                     000585
  1144   // "Info" object                                                                                                  000586
  1145   i_Info = RPDFG_newObject();                                                                                       000587
  1146   RPDFG_addDictEntry(i_Info:                                                                                        000588
  1147                      'Author':'(' + SplfAtr(1).UserName + ')');                                                     000589
  1148   RPDFG_addDictEntry(i_Info:                                                                                        000590
  1149                      'CreationDate':'(D:' + %CHAR(SplfAtr(1).CreateTimestamp:*ISO0) +                               000591
  1150                      '+00''00''' +                                                                                  000592
  1151                      ')');                                                                                          000593
  1152   RPDFG_addDictEntry(i_Info:'Creator':                                                                              000594
  1153                      '(RPDFGCVT https://github.com/AG1965/RPDFG)');                                                 000595
  1154   RPDFG_addDictEntry(i_Info:'Producer':                                                                             000596
  1155                      '(RPDFG Compiled (D:' + PSDS.CompileUDate + PSDS.CompileTime + ') IBMi)');                     000597
  1156   // /Title (No Title)                                                                                              000598
  1157   // /Subject (None)                                                                                                000599
  1158   // /Keywords (ClibPDF)                                                                                            000600
  1159   pdfObjRef_Info = RPDFG_ref(i_Info);                                                                               000601
  1160   RPDFG_serializeObject(i_Info);                                                                                    000602
  1161                                                                                                                     000603
  1162   // write all remaining objects (e.g. Pages objects that are not full and therefore not written ye                 000604
...   t)
  1163   RPDFG_serializeUnwrittenObjects();                                                                                000605
  1164                                                                                                                     000606
  1165   // xref                                                                                                           000607
  1166   i_Xref = RPDFG_newObject(RPDFG_SPECIAL_OBJECT.XREF);                                                              000608
  1167   RPDFG_serializeObject(i_Xref);                                                                                    000609
  1168                                                                                                                     000610
  1169   // trailer                                                                                                        000611
  1170   i_Trailer = RPDFG_newObject(RPDFG_SPECIAL_OBJECT.TRAILER);                                                        000612
  1171   RPDFG_addDictEntry(i_Trailer:'Size':%CHAR(RPDFG_getTotalNumberOfObjects() + 1));                                  000613
  1172   RPDFG_addDictEntry(i_Trailer:'Root':pdfObjRef_Root);                                                              000614
  1173   RPDFG_addDictEntry(i_Trailer:'Info':pdfObjRef_Info);                                                              000615
  1174   RPDFG_serializeObject(i_Trailer);                                                                                 000616
  1175                                                                                                                     000617
```

```
Line   <---------------------- Source Specifications ---------------------------------------------------->  Do  Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10  Num Date   Id  Number
  1176    // startxref + close PDF file                                                                                    000618
  1177    RPDFG_closePdfFile();                                                                                            000619
  1178                                                                                                                     000620
  1179 End-Proc;                                                                                                           000621
```

* * * * * E N D O F S O U R C E * * * * *

/ C o p y M e m b e r s

```
Line   Src  RPG name   <-------- External name -------> CCSID  <- Last change ->

Number Id              Library    File       Member            Date     Time
   103    1 RPDFG.RPG+ ./QSRVCPY/RPDFG.RPGLEINC          1208  26-04-01 10:48:01
   377    2 DIFS.RPGL+ ./QSYSCPY/DIFS.RPGLEINC           1208  26-03-27 08:21:08
   544    3 DERRNO.RP+ ./QSYSCPY/DERRNO.RPGLEINC         1208  26-03-27 19:38:00
   657    4 PERRNO.RP+ ./QSYSCPY/PERRNO.RPGLEINC         1208  26-03-26 09:18:32
```

* * * * * E N D O F / C O P Y M E M B E R S * * * * *

```
Line   <---------------------- Data Records ------------------------------------------------------------>      Change Src Seq
Number ....+....1....+....2....+....3....+....4....+....5....+....6....+....7....+....8....+....9....+...10      Date   Id  Number
                     C o m p i l e   T i m e   D a t a
*RNF0282 10   1162 000604  The record length of the output file is too small. The
                           record length should be at least 102.

    * * * * *  E N D  O F  C O M P I L E  T I M E  D A T A  * * * * *
```

M e s s a g e S u m m a r y

```
 Msg id  Sv Number Message text
*RNF0282 10      1 The record length of the output file is too small.
```

* * * * * E N D O F M E S S A G E S U M M A R Y * * * * *

F i n a l S u m m a r y

Message Totals:

```
  Information  (00) . . . . . . . :       0
  Warning      (10) . . . . . . . :       1
  Error        (20) . . . . . . . :       0
  Severe Error (30+)  . . . . . . :       0
  -------------------------------  -------
  Total . . . . . . . . . . . . . :       1
```

Source Totals:

```
  Records . . . . . . . . . . . . :    1179
  Specifications  . . . . . . . . :     813
  Data records  . . . . . . . . . :       0
  Comments  . . . . . . . . . . . :     351
```

* * * * * E N D O F F I N A L S U M M A R Y * * * * *

Diagnostic check of source is complete. Highest severity is 10.

* * * * * E N D O F C O M P I L A T I O N * * * * *